Weather Forecasting Model

Global Weather Repository - Analysis Report

Primary City: Buenos Aires

Data Summary

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======================================================================
  DATA SUMMARY REPORT
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 Shape: 130,393 rows × 41 columns

Column                                   Dtype                Non-Null
----------------------------------------------------------------------
country                                  str                   130,393
location_name                            str                   130,393
latitude                                 float64               130,393
longitude                                float64               130,393
timezone                                 str                   130,393
last_updated_epoch                       int64                 130,393
last_updated                             datetime64[us]        130,393
temperature_celsius                      float64               130,393
temperature_fahrenheit                   float64               130,393
condition_text                           str                   130,393
wind_mph                                 float64               130,393
wind_kph                                 float64               130,393
wind_degree                              int64                 130,393
wind_direction                           str                   130,393
pressure_mb                              float64               130,393
pressure_in                              float64               130,393
precip_mm                                float64               130,393
precip_in                                float64               130,393
humidity                                 int64                 130,393
cloud                                    int64                 130,393
feels_like_celsius                       float64               130,393
feels_like_fahrenheit                    float64               130,393
visibility_km                            float64               130,393
visibility_miles                         float64               130,393
uv_index                                 float64               130,393
gust_mph                                 float64               130,393
gust_kph                                 float64               130,393
air_quality_Carbon_Monoxide              float64               130,393
air_quality_Ozone                        float64               130,393
air_quality_Nitrogen_dioxide             float64               130,393
air_quality_Sulphur_dioxide              float64               130,393
air_quality_PM2.5                        float64               130,393
air_quality_PM10                         float64               130,393
air_quality_us-epa-index                 int64                 130,393
air_quality_gb-defra-index               int64                 130,393
sunrise                                  str                   130,393
sunset                                   str                   130,393
moonrise                                 str                   130,393
moonset                                  str                   130,393
moon_phase                               str                   130,393
moon_illumination                        int64                 130,393

======================================================================
  MISSING VALUES
======================================================================

 No missing values found in any column.


======================================================================
  NUMERICAL SUMMARY (describe)
======================================================================
                                 count                      mean              std                  min                  25%
latitude                      130393.0                  19.203898        24.420223                -41.3                 3.87
longitude                     130393.0                  21.975347        65.788585               -175.2              -6.8361
last_updated_epoch            130393.0          1744864197.998359  16741338.620674        1715849100.0        1730365200.0         1
last_updated                    130393  2025-04-17 06:39:06.082228              NaN  2024-05-16 01:45:00  2024-10-31 20:00:00  2025-04-
temperature_celsius           130393.0                  21.393049         9.690874                -29.8                 16.1
temperature_fahrenheit        130393.0                  70.509273        17.443417                -21.6                 61.0
wind_mph                      130393.0                   8.041106         7.291233                  2.2                  3.8
wind_kph                      130393.0                  12.944508        11.730955                  3.6                  6.1
wind_degree                   130393.0                 168.916192        103.59257                  1.0                 80.0
```

pressure_mb	130393.0	1014.098426	10.487106	947.0	1010.0
pressure_in	130393.0	29.945693	0.309649	27.96	29.83
precip_mm	130393.0	0.135171	0.564072	0.0	0.0
precip_in	130393.0	0.005125	0.022294	0.0	0.0
humidity	130393.0	66.42513	23.982923	2.0	50.0
cloud	130393.0	40.000537	34.098844	0.0	0.0
feels_like_celsius	130393.0	22.260845	11.585555	-36.7	16.1
feels_like_fahrenheit	130393.0	72.063764	20.85161	-34.0	61.0
visibility_km	130393.0	9.505747	2.68759	0.0	10.0
visibility_miles	130393.0	5.614926	1.679643	0.0	6.0
uv_index	130393.0	3.376691	3.556408	0.0	0.1
gust_mph	130393.0	11.365743	8.64565	2.2	6.4
gust_kph	130393.0	18.293315	13.913775	3.6	10.3
air_quality_Carbon_Monoxide	130393.0	468.394132	768.987483	-9999.0	207.0
air_quality_Ozone	130393.0	58.424157	30.950204	0.0	38.0
air_quality_Nitrogen_dioxide	130393.0	15.132711	23.965971	0.0	1.665
air_quality_Sulphur_dioxide	130393.0	10.518721	36.263727	-9999.0	1.1
air_quality_PM2.5	130393.0	24.44336	37.356829	0.168	7.15
air_quality_PM10	130393.0	48.920907	150.861815	-1848.15	10.0
air_quality_us-epa-index	130393.0	1.706411	0.945979	1.0	1.0
air_quality_gb-defra-index	130393.0	2.623607	2.459816	1.0	1.0
moon_illumination	130393.0	49.579625	35.069177	0.0	15.0

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======================================================================
  CATEGORICAL COLUMNS ? TOP VALUES
======================================================================

 Occurrences of country (211 unique)
----------------------------------------
  Bulgaria                          1,545  (1.2%)
  Indonesia                         1,341  (1.0%)
  Thailand                          1,340  (1.0%)
  Turkey                            1,336  (1.0%)
  Sudan                             1,336  (1.0%)
  Bolivia                           1,329  (1.0%)
  Iran                              1,304  (1.0%)
  Belgium                           1,266  (1.0%)
  Madagascar                        1,128  (0.9%)
  Vietnam                           1,112  (0.9%)

 Occurrences of location_name (257 unique)
----------------------------------------
  Kabul                               671  (0.5%)
  Bujumbura                           671  (0.5%)
  N'djamena                           671  (0.5%)
  Malabo                              671  (0.5%)
  Asmara                              671  (0.5%)
  Suva                                671  (0.5%)
  Accra                               671  (0.5%)
  Vatican City                        671  (0.5%)
  Baghdad                             671  (0.5%)
  Tokyo                               671  (0.5%)

 Occurrences of timezone (199 unique)
----------------------------------------
  Asia/Bangkok                      2,421  (1.9%)
  Europe/Sofia                      1,509  (1.2%)
  Europe/Rome                       1,447  (1.1%)
  America/La_Paz                    1,354  (1.0%)
  Asia/Jakarta                      1,341  (1.0%)
  Europe/Istanbul                   1,338  (1.0%)
  Africa/Khartoum                   1,336  (1.0%)
  Asia/Tehran                       1,304  (1.0%)
  Europe/Brussels                   1,267  (1.0%)
  Indian/Antananarivo               1,128  (0.9%)

 Occurrences of condition_text (49 unique)
----------------------------------------
  Partly cloudy                    38,053  (29.2%)
  Sunny                            37,644  (28.9%)
  Patchy rain nearby               10,761  (8.3%)
  Partly Cloudy                     7,923  (6.1%)
  Overcast                          7,011  (5.4%)
  Clear                             6,746  (5.2%)
  Mist                              5,462  (4.2%)
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```
  Light rain                        4,080  (3.1%)
  Light rain shower                 2,999  (2.3%)
  Fog                               1,621  (1.2%)

 Occurrences of wind_direction (16 unique)
----------------------------------------
  E                                12,352  (9.5%)
  ENE                              10,827  (8.3%)
  ESE                               9,019  (6.9%)
  N                                 8,688  (6.7%)
  NNE                               8,554  (6.6%)
  S                                 7,992  (6.1%)
  WSW                               7,717  (5.9%)
  SSW                               7,709  (5.9%)
  SW                                7,685  (5.9%)
  SSE                               7,639  (5.9%)

 Occurrences of sunrise (495 unique)
----------------------------------------
  06:10 AM                          1,347  (1.0%)
  05:57 AM                          1,345  (1.0%)
  05:56 AM                          1,324  (1.0%)
  06:11 AM                          1,319  (1.0%)
  06:17 AM                          1,313  (1.0%)
  05:55 AM                          1,312  (1.0%)
  06:21 AM                          1,299  (1.0%)
  06:24 AM                          1,293  (1.0%)
  05:54 AM                          1,281  (1.0%)
  05:58 AM                          1,278  (1.0%)

 Occurrences of sunset (547 unique)
----------------------------------------
  06:02 PM                          1,287  (1.0%)
  06:03 PM                          1,210  (0.9%)
  06:28 PM                          1,167  (0.9%)
  06:11 PM                          1,148  (0.9%)
  06:13 PM                          1,145  (0.9%)
  06:17 PM                          1,124  (0.9%)
  06:04 PM                          1,122  (0.9%)
  06:06 PM                          1,118  (0.9%)
  06:21 PM                          1,116  (0.9%)
  06:29 PM                          1,116  (0.9%)

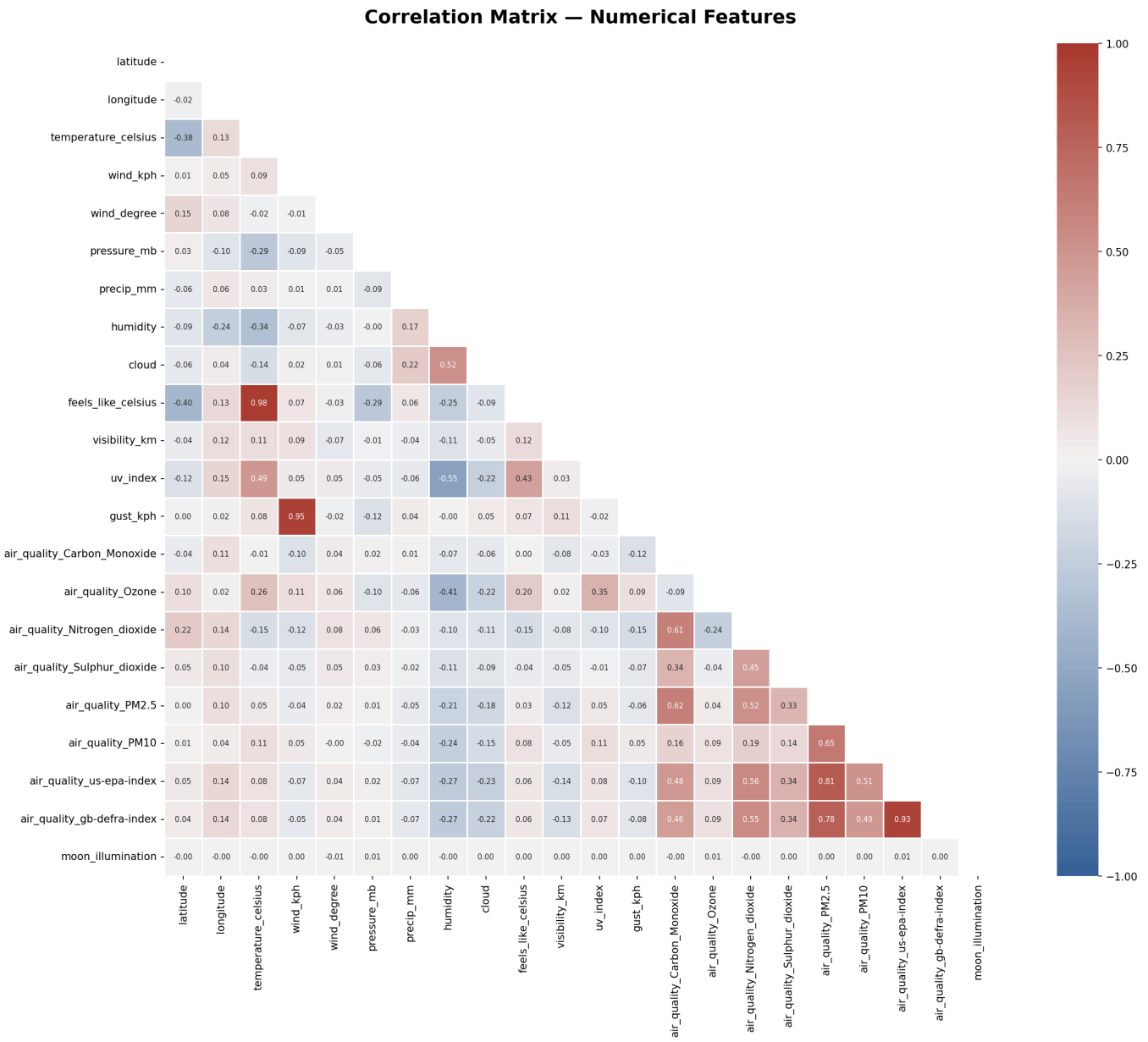
 Occurrences of moonrise (1444 unique)
----------------------------------------
  No moonrise                       4,448  (3.4%)
  12:00 AM                            245  (0.2%)
  01:05 PM                            127  (0.1%)
  09:27 AM                            118  (0.1%)
  12:51 AM                            116  (0.1%)
  05:52 AM                            116  (0.1%)
  11:45 AM                            116  (0.1%)
  02:05 PM                            115  (0.1%)
  01:54 PM                            113  (0.1%)
  06:15 PM                            113  (0.1%)

 Occurrences of moonset (1444 unique)
----------------------------------------
  No moonset                        4,276  (3.3%)
  12:00 AM                            252  (0.2%)
  10:33 AM                            123  (0.1%)
  05:53 PM                            121  (0.1%)
  01:08 AM                            119  (0.1%)
  12:01 AM                            119  (0.1%)
  01:55 AM                            118  (0.1%)
  01:53 AM                            116  (0.1%)
  02:03 AM                            114  (0.1%)
  09:31 AM                            114  (0.1%)

 Occurrences of moon_phase (8 unique)
----------------------------------------
  Waning Crescent                  28,697  (22.0%)
  Waxing Gibbous                   28,094  (21.5%)
  Waxing Crescent                  27,903  (21.4%)
  Waning Gibbous                   27,135  (20.8%)
```

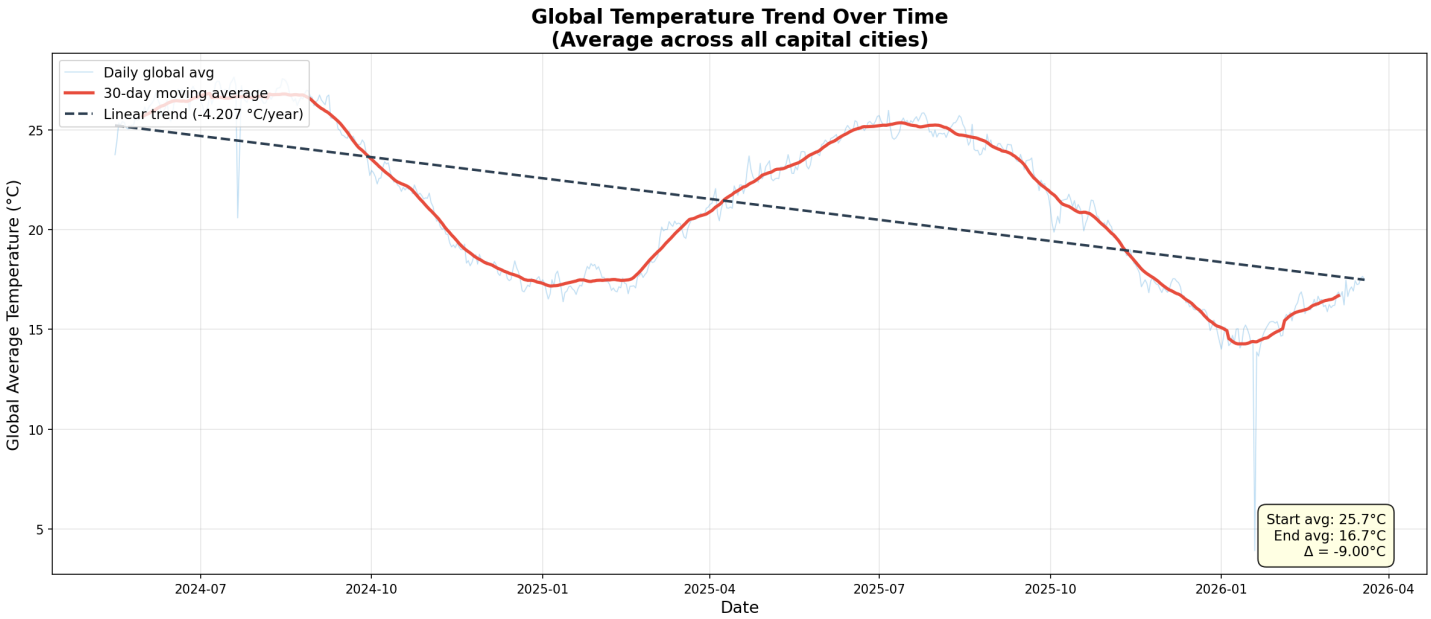
Full Moon	4,715	(3.6%)
Last Quarter	4,702	(3.6%)
New Moon	4,676	(3.6%)
First Quarter	4,471	(3.4%)

Correlation Heatmap



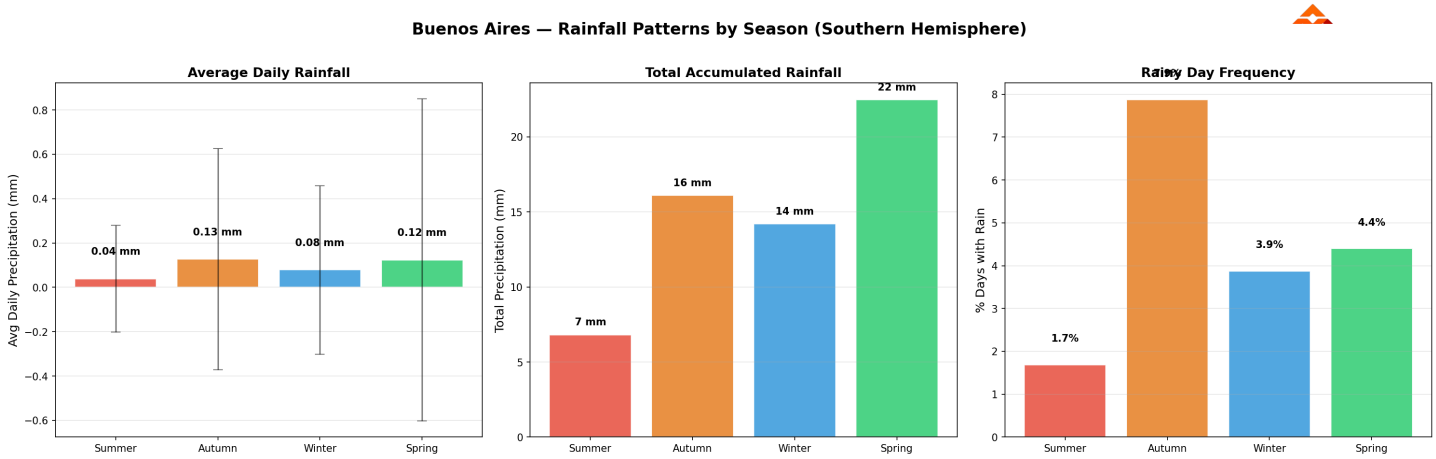
This heatmap highlights relationships between weather variables. Strong correlations (e.g., temperature and feels-like temperature) appear in dark red, while negative correlations appear in dark blue. It helps identify which meteorological features are heavily interdependent, allowing us to create more insightful visualizations.

Global Temperature Trend



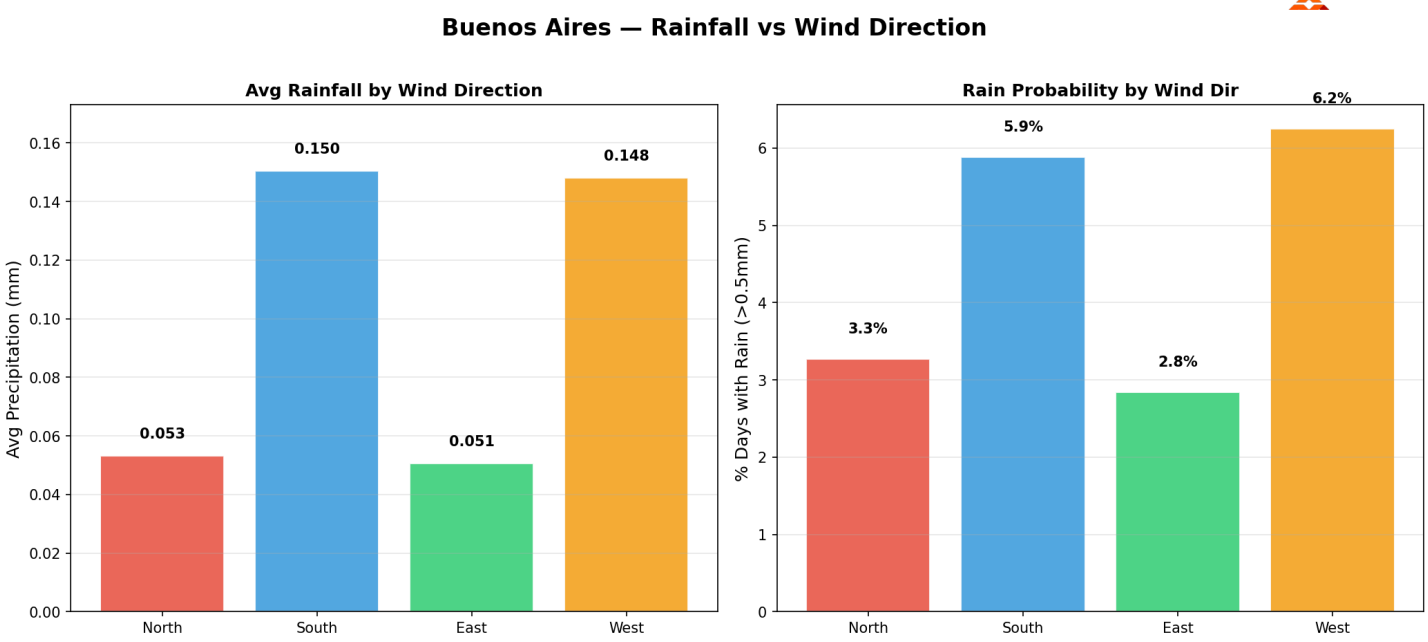
This line chart shows the average global temperature over the available dataset period. While long-term upward trajectories generally point toward the effects of global warming, the sharp declining trend seen here is contrary to expectations. This anomaly is explained by two main factors. First, global temperature trends are normally measured in decades rather than a span of a few years. Second, the dataset averages temperatures across all capital cities; because the vast majority of capital cities are located in the Northern Hemisphere, the data is heavily skewed. Consequently, instead of a true long-term climate shift, the chart primarily captures a natural seasonal transition from a Northern Hemisphere summer peak to a winter trough.

BA - Rainfall by Season



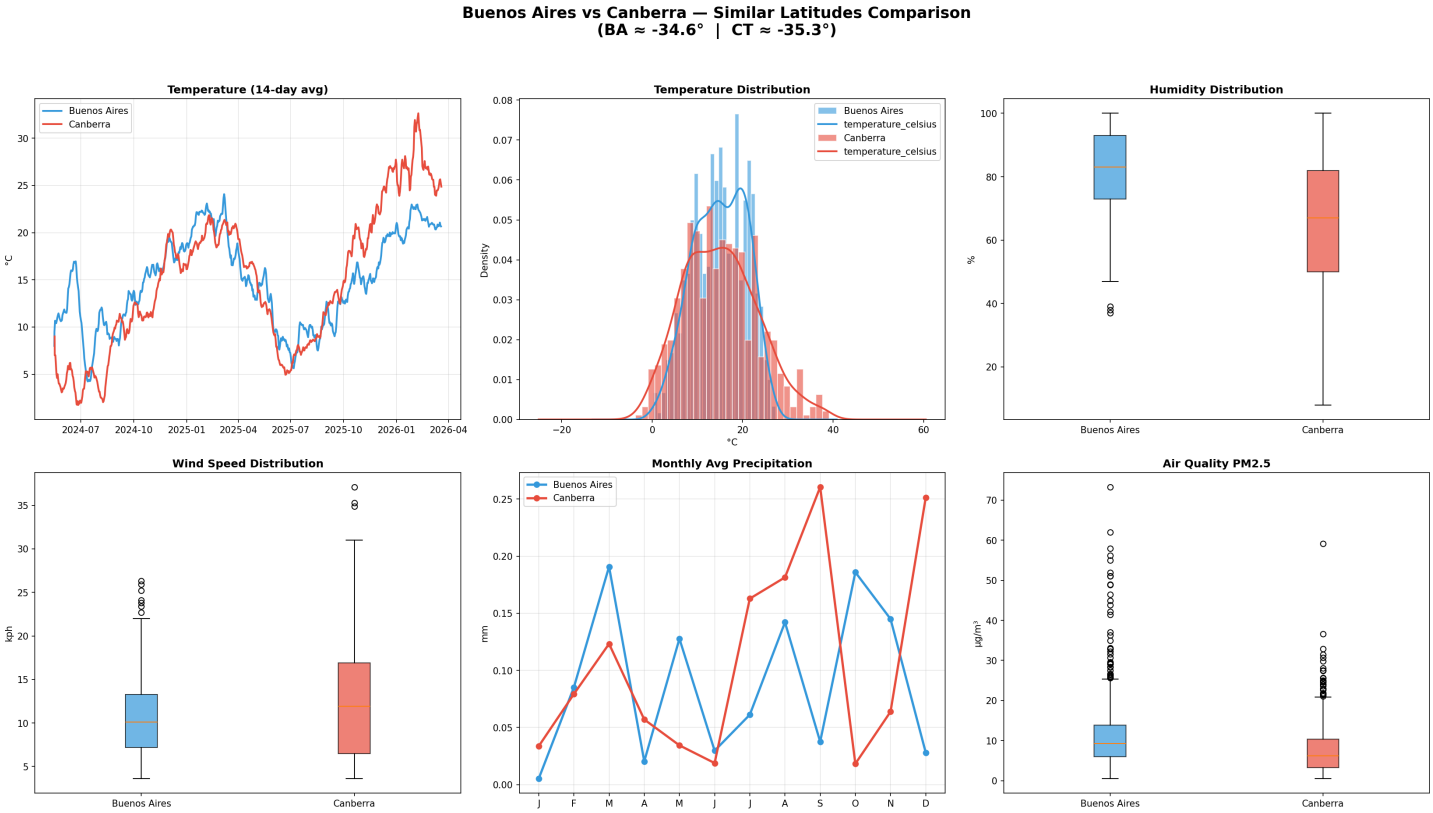
This series of charts breaks down rainfall in Buenos Aires by season. It highlights the city's wettest and driest periods, which are largely driven by regional atmospheric patterns and seasonal shifts in humidity. Interestingly, the data shows that spring, rather than summer, is the season with the highest accumulated rainfall.

BA - Rain vs Wind Direction



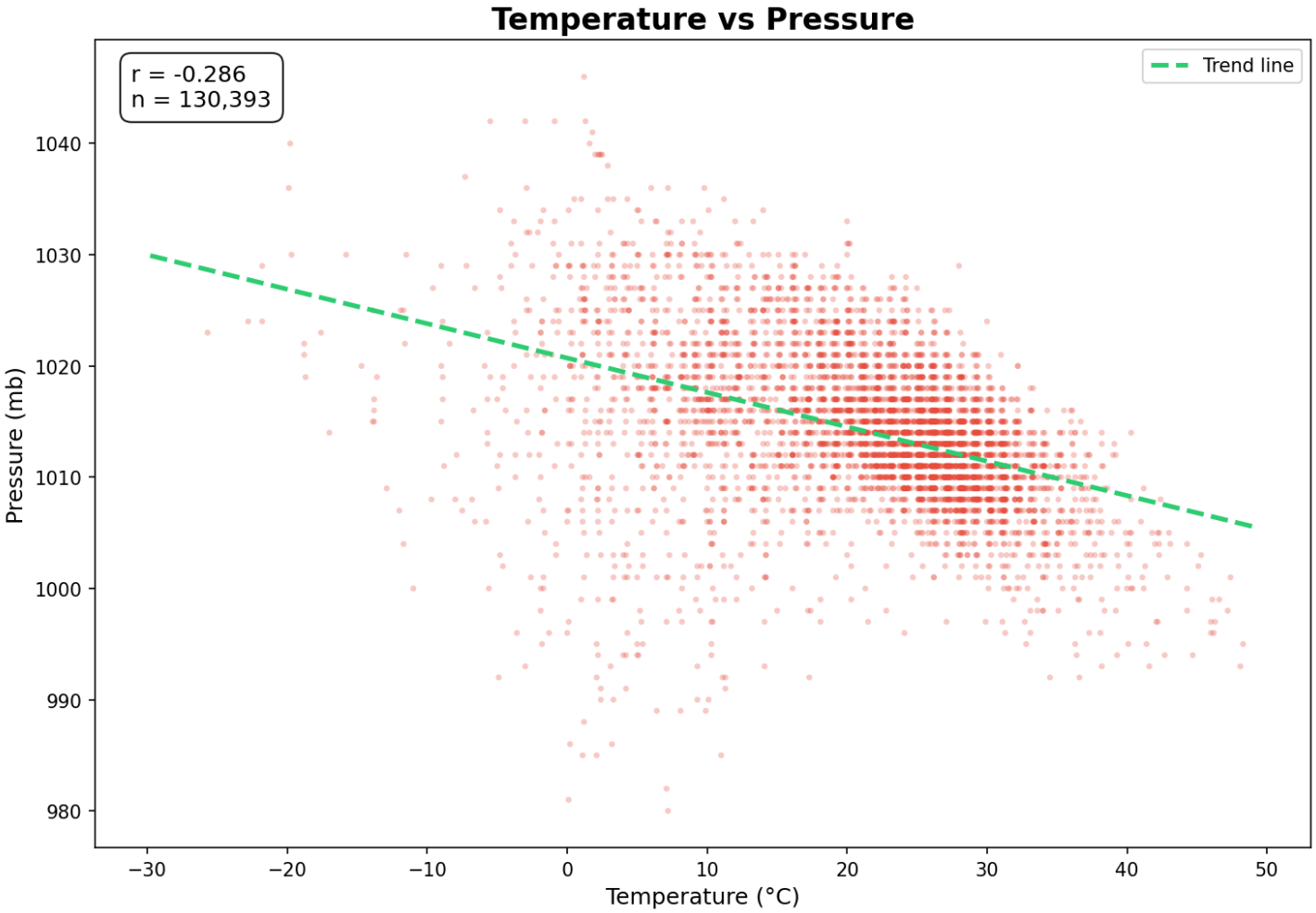
This visualizes the relationship between wind direction and precipitation in Buenos Aires. While the data clearly shows that South and West winds are highly correlated with rainfall in Buenos Aires, this is driven by local atmospheric dynamics rather than the winds themselves acting as the primary source of moisture. Typically, the region experiences a buildup of warm, highly humid air brought by Northern winds. When a cold, dense front from the South or Southwest (a regional phenomenon known as the Pampero) moves in, it collides with this unstable air mass. The heavier cold air forces the warm moisture upward rapidly, causing sudden condensation and triggering heavy thunderstorms. Therefore, the high precipitation recorded during South and West winds captures the exact moment these cold fronts break the accumulated humidity, acting as a climatic trigger rather than a carrier of rain.

BA vs Canberra Comparison



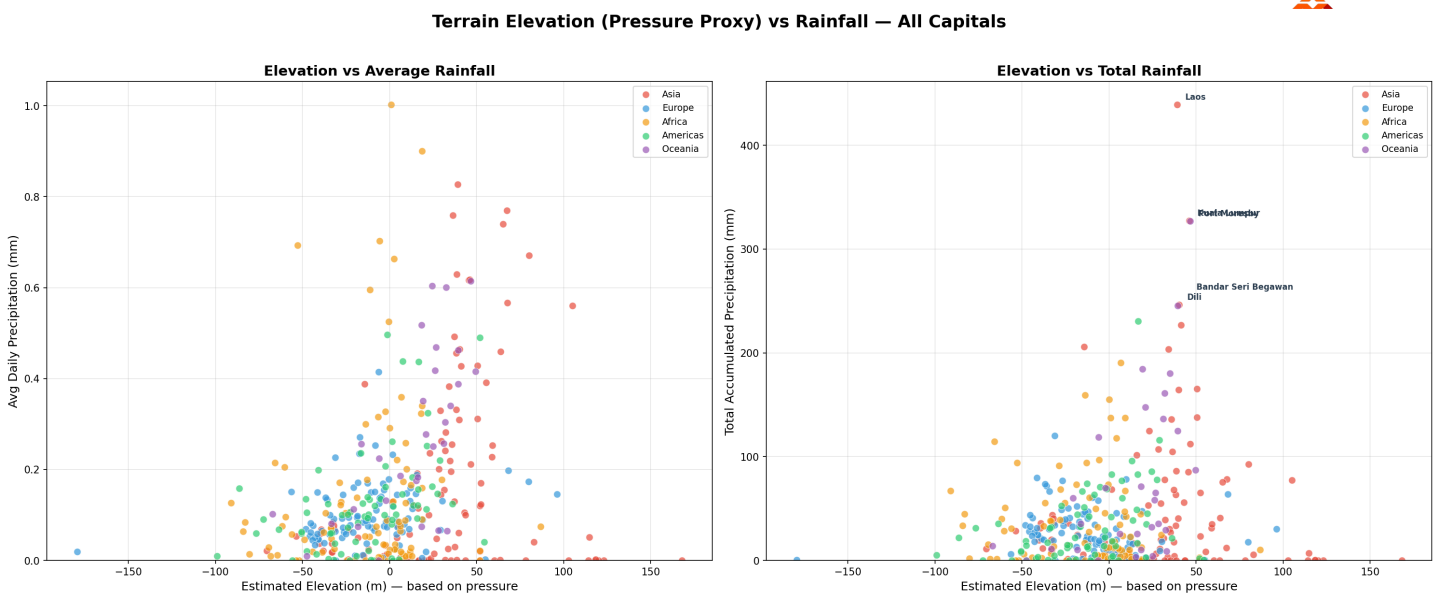
This dashboard compares the climate profiles of Buenos Aires and Canberra, two cities selected for their nearly identical latitudes (~35° South). While their shared latitude dictates perfectly synchronized seasonal cycles, the data reveals stark contrasts driven by their geographical contexts. Canberra exhibits a classic continental climate, characterized by wider temperature extremes?hotter summer peaks and colder winter valleys?and significantly lower, highly variable humidity. In contrast, Buenos Aires displays a narrower temperature distribution and consistently high humidity, with a median hovering near 80%. This comparison elegantly highlights how coastal proximity and oceanic thermal buffering (in Buenos Aires) versus an inland, elevated location (in Canberra) can drastically alter a region's climate despite receiving the exact same solar radiation.

Temperature vs Pressure



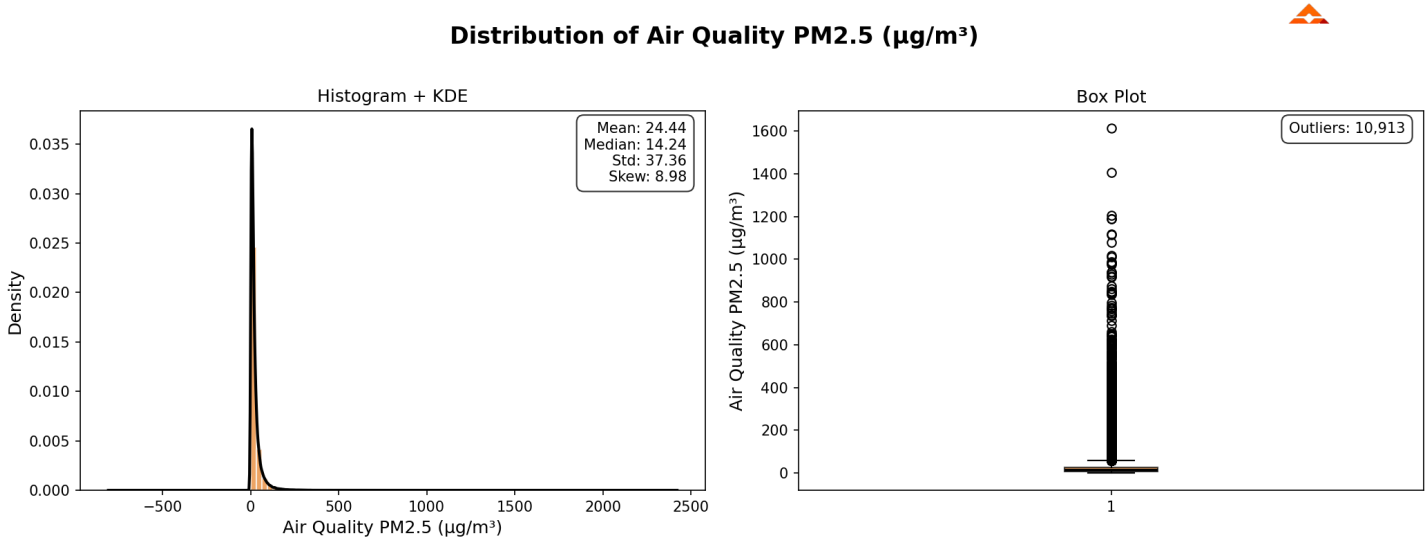
This scatter plot illustrates the relationship between temperature and atmospheric pressure across a substantial dataset of over 130,000 observations. The negative Pearson correlation coefficient (r = -0.286) and the downward trend line confirm the physical principle that warmer air tends to be less dense, resulting in lower atmospheric pressure. However, the wide dispersion of data points and the relatively weak correlation value indicate that temperature is just one of many variables influencing pressure. Other geographic and meteorological factors strongly contribute to atmospheric pressure variations at any given temperature.

Elevation vs Rainfall



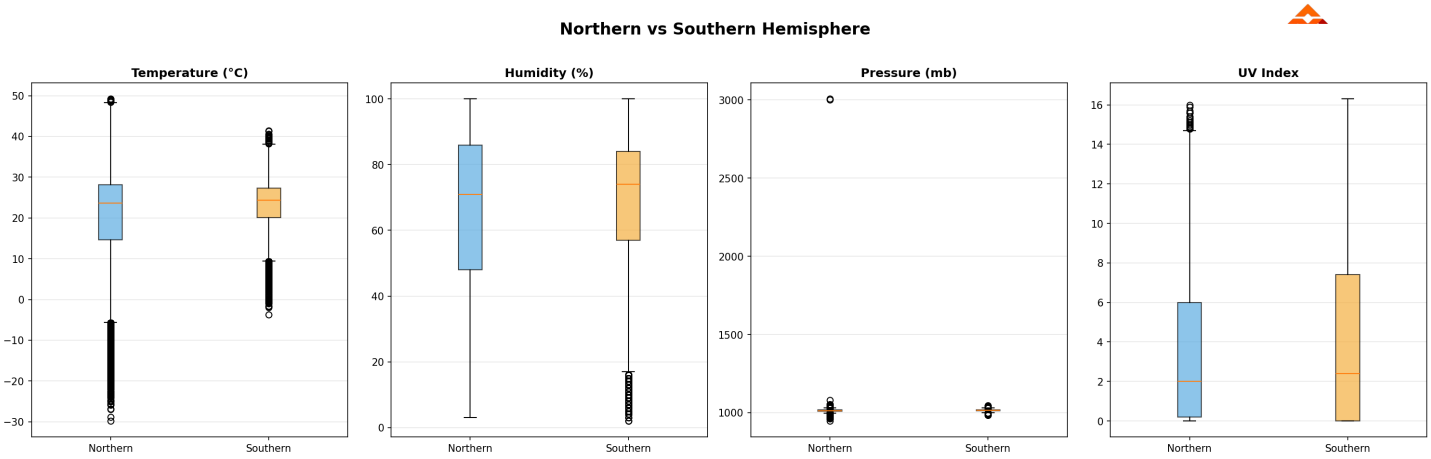
While theoretical meteorology dictates that elevation influences precipitation, this scatter plot reveals that across global capital cities, elevation is not the sole driver of rainfall. Instead of a strong linear correlation, the data forms a dense central cluster. However, the visualization effectively highlights distinct outliers?such as Laos, Port Moresby, and Bandar Seri Begawan (predominantly in Asia and Oceania)?where extreme accumulated rainfall occurs at moderate estimated elevations. This suggests that while topography plays a localized role, macro-climatic factors like tropical monsoon seasons are the primary drivers of the world's most extreme precipitation levels.

Air Quality PM2.5 Distribution



This visualization illustrates the highly skewed global distribution of PM2.5 air pollution. As evidenced by the extreme positive skewness (8.98) and a mean (24.44) that is significantly pulled higher than the median (14.24), the vast majority of global records cluster within lower, healthier air quality ranges. However, the box plot reveals nearly 11,000 severe outliers, creating a massive 'long tail' in the dataset. These extreme values do not represent standard daily fluctuations, but rather acute environmental events?such as intense industrial emissions, seasonal crop burning, or geographic basins trapping smog. Ultimately, this demonstrates that while the 'typical' global air quality might appear moderate, the distribution is heavily punctuated by localized, severe pollution crises.

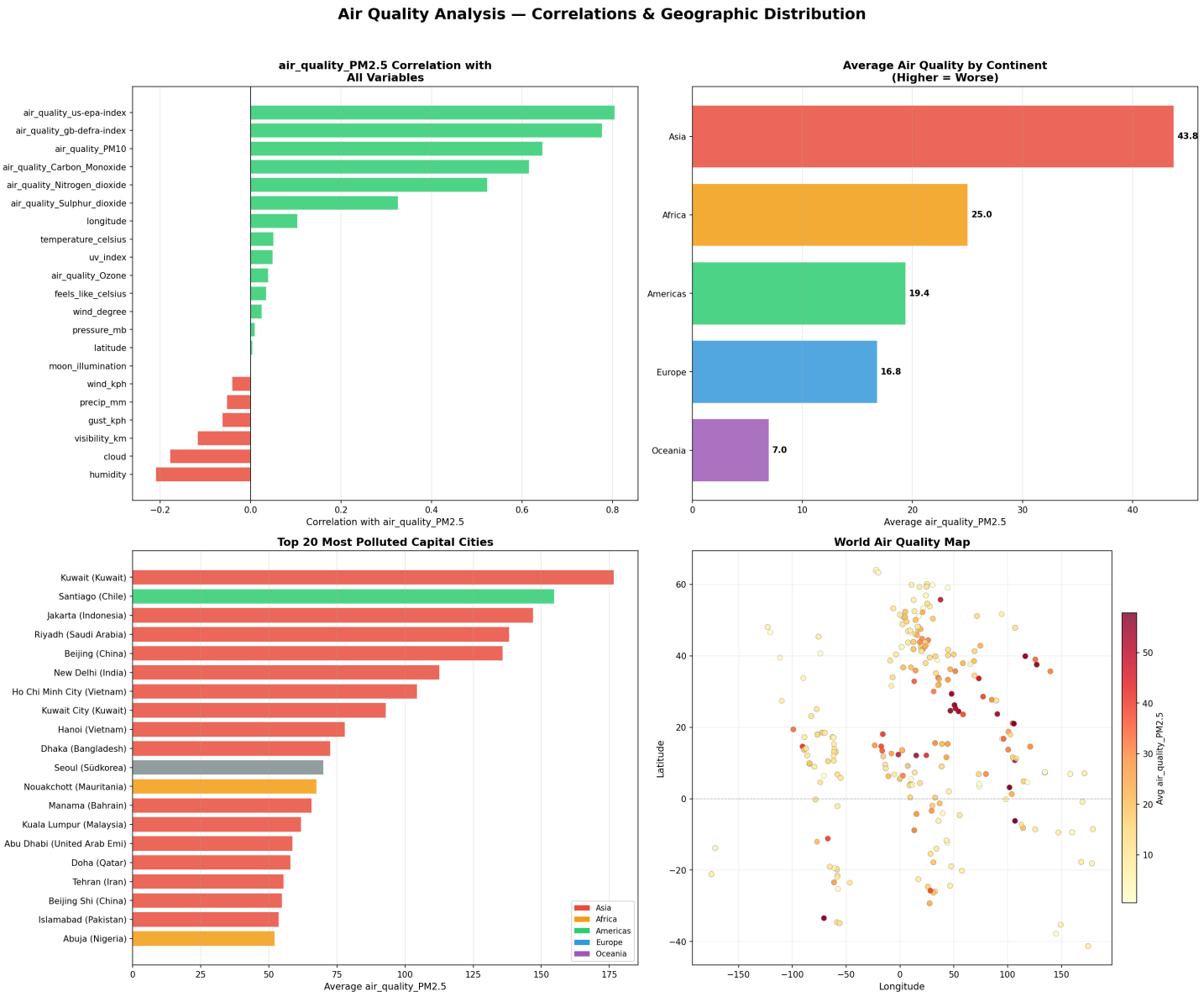
Hemisphere Comparison



This dashboard highlights the structural climatic differences between the Northern and Southern Hemispheres. As the box plots demonstrate, the Northern Hemisphere exhibits extreme temperature volatility?evidenced by a much wider interquartile range and severe cold outliers dropping below -20°C. This is driven by its vast landmasses, which heat and cool rapidly. Conversely, the Southern Hemisphere shows a much tighter, more stable temperature distribution, reflecting the thermal buffering effect of its dominant oceans.

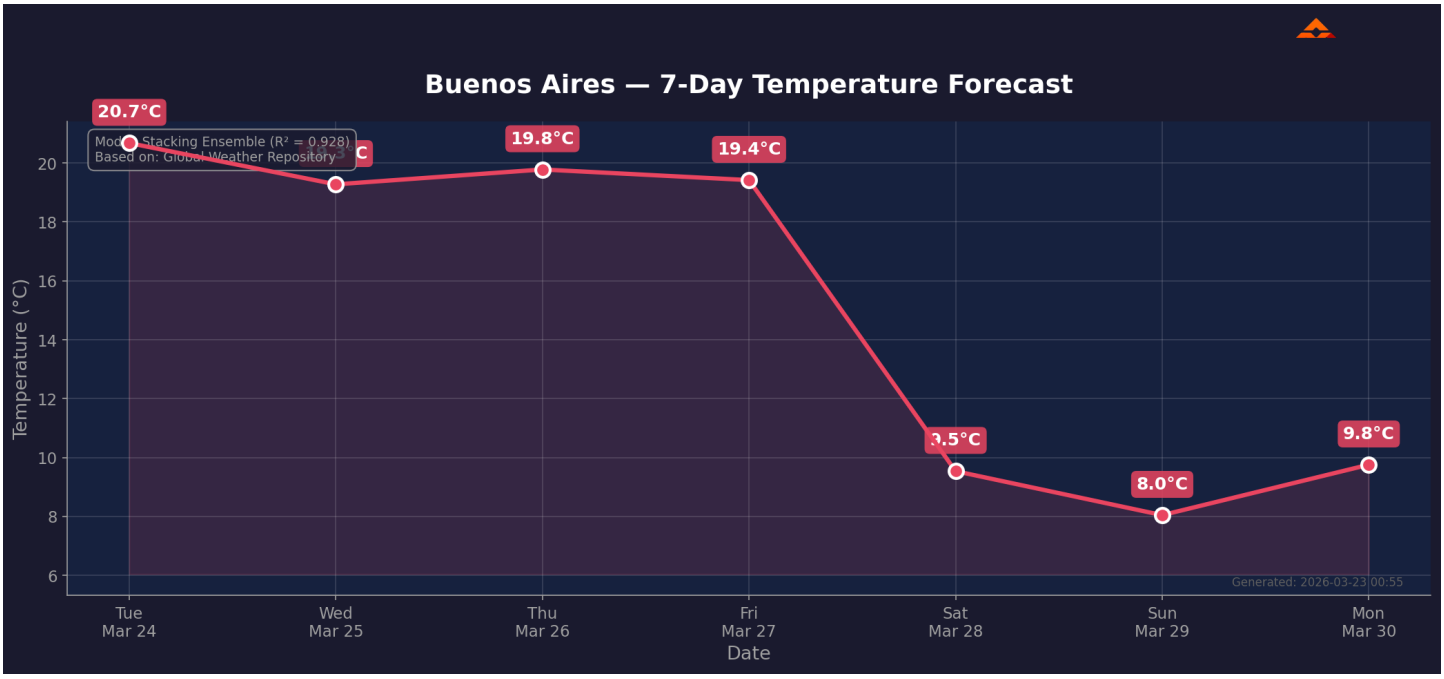
Furthermore, this visualization serves as a crucial data quality check: the Pressure chart reveals an impossible outlier in the Northern Hemisphere at approximately 3000 mb (standard sea-level pressure is ~1013 mb). This clearly indicates a sensor error or a corrupt data point in the raw dataset that requires cleaning.

Air Quality Geographic Analysis



This comprehensive dashboard provides a global overview of air quality, specifically focusing on PM2.5 particulate matter. The geographic analysis starkly identifies Asia as the most severely impacted continent, averaging a PM2.5 level of 43.8. This is heavily reflected in the 'Top 20 Most Polluted Capitals' list, which is overwhelmingly dominated by Asian and Middle Eastern cities. However, a significant anomaly emerges in the Americas: Santiago (Chile) ranks as the second most polluted capital globally, likely due to its unique topography (a valley surrounded by mountains) that traps smog. Furthermore, the correlation analysis confirms that PM2.5 concentration is heavily tied to other chemical pollutants (like Carbon Monoxide), while negatively correlated with humidity and cloud cover, indicating that wet, rainy conditions act as a natural cleanser for atmospheric particulate matter.

7-Day Temperature Forecast ? Buenos Aires



This chart presents the predicted temperature for Buenos Aires over the next 7 days, generated using the Stacking Ensemble model (R2 = 0.928). The model combines Random Forest, Gradient Boosting, XGBoost, and LightGBM as base learners with a Ridge meta-learner. Predictions are based on the most recent weather observations for Buenos Aires from the dataset, with small day-to-day perturbations applied to simulate natural variability. Note: this is a model-based estimation, not a live meteorological forecast.